

ThisFiTT V1 — Final Rule Book

Developer Handoff | v5 (Consistency Pass)

v5 changes (vs v4):

- Warm-up block spec corrected: **2–4 movements** (was 4–6) — R29.
- Block 4 rep-range rules now use the actual **Load type** enum (heavy strength, power, hypertrophy, endurance, speed strength, strength speed, prehab, mobility) — R61–R66.
- Within-session rep behavior clarified: ascending Load types use **within-session rep descent** (e.g., hypertrophy 12/10/8); endurance and tendon isometric stay constant — R161.
- "Warm-up sets" inside R71 renamed to **ramp sets** to avoid collision with the session-level Warm-up block in R29.
- Week-1 RPE rule (R76) explicitly overrides R74 across all athlete levels.
- Rest tiebreaker added (R164): when multiple rest rules apply, use the longest required value.
- Cross-references in Block 1 cleaned up (R5, R9, R18 now reference correct rule numbers).
- R71 no longer duplicates Block 6 ramp logic — Block 4 governs sets/rest, Block 6 governs per-set load.
- Taper rule (formerly R147) **removed** for V1 — no competition-date trigger exists in onboarding. Will return when onboarding supports it.

Rule 0 — Precedence

When two rules conflict, the higher-priority block wins:

Block 1 (Safety & Pain Overrides) > Block 4 (CNS, Recovery, Volume) > Block 5 (Phase, Progression, Templates) > Block 2 (Session Structure) > Block 3 (Exercise Selection).

Block 6 (Per-Set Load Prescription) is a display layer applied AFTER Blocks 1–5 have filtered, ordered, and selected exercises.

Block 1 — Safety & Pain Overrides (RUNS FIRST)

- R1.** IF athlete age is under 18 THEN restrict maximal loading above 90% effort.
- R2.** IF athlete fails the medical clearance gate at onboarding THEN do not generate a training plan; show medical clearance screen.
- R3.** IF athlete reports knee pain, severity = severe THEN exclude all knee-stress exercises and exclude all plyometrics; show medical clearance screen.
- R4.** IF athlete reports knee pain, severity = moderate THEN exclude exercises with knee_stress = high, reduce high-impact plyometrics, and include knee-area prehab work weekly.
- R5.** IF athlete reports knee pain, severity = mild AND type = chronic THEN allow most exercises but cap knee-stress exercises at moderate, cap intensity at 80%, and include knee tendon protocol per R79.
- R6.** IF athlete reports knee pain, severity = mild AND type = acute THEN treat as moderate severity for 2 weeks, then re-prompt user to reassess.
- R7.** IF athlete reports shin splints, severity = severe THEN exclude all repetitive impact work; show medical clearance screen.
- R8.** IF athlete reports shin splints, severity = moderate THEN reduce repetitive pogo and sprint volume by at least 50% and exclude high-impact plyometrics.
- R9.** IF athlete reports shin splints, severity = mild THEN reduce pogo and sprint volume by 25% and include calf tendon work per R78.
- R10.** IF athlete reports back pain, severity = severe THEN exclude all axial loading and all spine-stress exercises; show medical clearance screen.

- R11.** IF athlete reports back pain, severity = moderate THEN exclude exercises with spine_stress = high and remove heavy axial loading lifts.
- R12.** IF athlete reports back pain, severity = mild THEN allow moderate axial loading but cap intensity at 80% and include core stability work weekly.
- R13.** IF athlete reports shoulder pain, severity = severe THEN exclude all overhead and pressing work; show medical clearance screen.
- R14.** IF athlete reports shoulder pain, severity = moderate THEN exclude exercises with shoulder_stress = high and exclude overhead pressing.
- R15.** IF athlete reports shoulder pain, severity = mild THEN cap pressing intensity at 80% and include scapular stability work weekly.
- R16.** IF athlete reports tendon or Achilles pain, severity = severe THEN exclude all plyometrics; show medical clearance screen.
- R17.** IF athlete reports tendon or Achilles pain, severity = moderate THEN exclude high-intensity plyometrics, prioritize isometric loading, and progress only per rule 100.
- R18.** IF athlete reports tendon or Achilles pain, severity = mild THEN reduce plyometric intensity by one tier and include tendon work per R78–R80 (calf, knee, Achilles as relevant).
- R19.** IF exercise.equipment is NOT available to the athlete THEN exclude exercise.
- R20.** IF exercise.joint_stress_knee = high AND athlete reports knee pain (any severity) THEN exclude exercise.
- R21.** IF exercise.joint_stress_spine = high AND athlete reports back pain (any severity) THEN exclude exercise.
- R22.** IF exercise.joint_stress_shoulder = high AND athlete reports shoulder pain (any severity) THEN exclude exercise.
- R23.** IF a session contains 2+ exercises with knee_stress = high THEN no additional knee_stress = high exercises may be added to that session, AND the next lower-body exercise scheduled must have knee_stress ≤ low.
- R24.** IF a session contains 2+ exercises with spine_stress = high THEN no additional axial-loaded exercises may be added to that session, AND the next training day must not begin with a spine_stress = high exercise.
- R25.** IF a session contains 2+ exercises with shoulder_stress = high THEN no additional pressing or overhead exercises may be added to that session.
- R26.** IF a training week contains 4+ exercises with [joint]_stress = high for the same joint THEN the next week must reduce that joint's high-stress exercises by at least 50%.
- R27.** IF athlete reports soreness ≥ 7/10 THEN reduce next workout intensity by one tier (e.g., 4–6 reps becomes 6–8 reps; load drops 10%).
- R28.** IF an exercise violates any rule in Block 1 THEN exclude exercise, regardless of session goal.

Block 2 — Session Structure & Ordering

Every session MUST contain a Warm-up block (placed first) and a Cool-down block (placed last, unless Recovery is present and replaces Cool-down per rule 33). Primer, Accessory, Conditioning, and Recovery blocks are optional and included based on day type, goal, and constraints. When optional blocks are included, they must follow the placement order defined in rules 30–34.

- R29.** IF a session begins THEN the first block must be a Warm-up consisting of mobility and activation work (typically 2–4 movements, 5–10 minutes total).
- R30.** IF session contains a Primer block THEN it comes after Warm-up and before the primary block.
- R31.** IF session contains an Accessory block THEN it comes after the primary block.

- R32.** IF session contains a Conditioning block THEN it comes after strength or power work.
- R33.** IF session contains a Cool-down AND Recovery is present THEN Cool-down must immediately precede Recovery, and Recovery is the final block.
- R34.** IF session contains a Cool-down AND Recovery is absent THEN Cool-down must be the final block.
- R35.** IF day type $\in$ {Full Body Strength, Lower Strength, Upper Strength} AND primary strength movement is complete THEN next exercise category must be secondary strength movement.
- R36.** IF day type $\in$ {Full Body Strength, Lower Strength, Upper Strength} AND secondary strength movement is complete THEN next exercise category must be accessory exercises.
- R37.** IF day type $\in$ {Full Body Strength, Lower Strength, Upper Strength, Upper + Mobility} AND accessory work is complete THEN next exercise category must be injury prevention or core.
- R38.** IF day type $\in$ {Power + Speed, Power, Speed} THEN exercise order is: Warm-up $\rightarrow$ Primer $\rightarrow$ Power (jumps, throws, sprints) $\rightarrow$ Accessory (if included) $\rightarrow$ Cool-down.
- R39.** IF day type = Recovery or Mobility THEN exercise order is: Warm-up $\rightarrow$ Mobility and soft tissue $\rightarrow$ Low-intensity activation $\rightarrow$ Recovery (final block).
- R40.** IF day type = Lower Unilateral or Volume THEN exercise order is: Warm-up $\rightarrow$ Primer $\rightarrow$ Unilateral lower work (primary) $\rightarrow$ Posterior chain or core accessory $\rightarrow$ Cool-down.
- R41.** IF a sprint session occurs THEN sprint work is performed before heavy strength work.
- R42.** IF exercise.CNS_cost = high THEN place exercise early in the session.
- R43.** IF exercise requires high technical skill THEN place exercise early in the session.
- R44.** IF a session contains ≥ 2 high-CNS exercises THEN exclude additional high-CNS exercises from that session.

Required Blocks by Day Type

Day Type	Required Blocks	Optional Blocks
Full Body Strength / Lower Strength / Upper Strength	Warm-up, Primer, Main Lift, Accessory, Cool-down	Conditioning, Recovery
Power + Speed / Power / Speed	Warm-up, Primer, Power, Cool-down	Accessory, Conditioning, Recovery
Upper + Mobility	Warm-up, Main Lift or Accessory, Cool-down	Primer, Recovery
Recovery / Mobility	Warm-up, Recovery	Cool-down folded into Recovery
Lower Unilateral / Volume	Warm-up, Accessory, Cool-down	Primer, Conditioning, Recovery

Block 3 — Exercise Selection & Equipment

- R45.** IF athlete has weight room access THEN allow barbell compound lifts.
- R46.** IF athlete has no weight room access THEN exclude exercises requiring barbell equipment.
- R47.** IF session contains a hinge pattern THEN include a knee-dominant accessory.
- R48.** IF session contains a push movement THEN include a pull movement.
- R49.** IF session contains a bilateral exercise THEN include a unilateral exercise.

- R50.** IF session contains a sagittal-plane movement THEN include a frontal-plane or transverse-plane movement.
- R51.** IF plyometric.contact_time = fast THEN classify exercise as reactive plyometric.
- R52.** IF plyometric.amplitude = high THEN limit volume to ≤ 20 contacts.
- R53.** IF athlete level = beginner THEN exercise.stability_demand must be $\leq$ moderate.
- R54.** IF athlete level = intermediate THEN exercise.stability_demand may be $\leq$ high.
- R55.** IF athlete level = advanced THEN allow all stability levels.
- R56.** IF athlete level = beginner THEN exclude advanced-level exercises.
- R57.** IF an accessory exercise is repeated for 3 consecutive weeks THEN rotate exercise. Main lifts (primary strength movements) are NOT rotated within a training block; they progress in load per Block 6 and only rotate at block transitions every 4–6 weeks.
- R58.** stability_demand is a required database attribute on every exercise, tagged as: low (machine-based, fully supported), moderate (bilateral free weight on stable surface), or high (unilateral, unstable surface, or complex coordination). Exercises without this tag cannot be selected by the generator.
- R59.** Joint stress attributes (knee_stress, spine_stress, shoulder_stress) are required database attributes on every exercise, tagged as: low / moderate / high. Exercises without these tags cannot be selected by the generator.
- R60.** CNS_cost and fatigue_cost are required database attributes on every exercise, tagged as: low / moderate / high.

Block 4 — Volume, Reps, Sets, Rest, Plyo Contacts

*Block 4 governs how many sets and reps are prescribed and how much rest is taken. **Per-set load distribution (the actual %1RM for each individual set) is governed by Block 6.** The old single-load-per-exercise model in prior versions of R69–R71 has been replaced; see Block 6 for the per-set scheme.*

- R61.** IF Load type = heavy strength THEN reps fall within 1–5 (range is the within-session descent band; see R161 for distribution).
- R62.** IF Load type = power (Olympic lifts, jump squats, med ball throws) THEN reps fall within 1–3 with maximal intent (within-session descent per R161).
- R63.** IF Load type = hypertrophy THEN reps fall within 8–12 (within-session descent per R161).
- R64.** IF Load type = endurance THEN reps = 12–15 reps held constant across all sets (straight scheme; no descent).
- R65.** IF Load type = speed strength OR strength speed THEN reps fall within 3–5 (within-session descent per R161).
- R66.** IF Load type = prehab AND exercise is a tendon isometric (Spanish squat, isometric squat, calf isometric) THEN duration = 30–45 seconds per hold, held constant across all sets within a session (see R161).
- R67.** IF athlete training experience = beginner THEN limit heavy strength lifts to 2 per session; IF athlete training experience = advanced THEN allow up to 3 strength movements per session.
- R68.** IF athlete performs a hip hinge movement THEN accessory work must include posterior chain support
- R69–R71 (Updated v4):** Set count determines rest and the number of ascending load steps. Per-set load percentages are assigned by Block 6 based on the exercise's Load type field.
- R69.** IF sets = 2–3 THEN rest = 120–150 seconds. Reps follow the Load type rep range (R61–R66). Per-set load distribution is governed by Block 6.

- R70.** IF sets = 4–5 THEN rest = 120–180 seconds. Reps follow the Load type rep range (R61–R66). Per-set load distribution is governed by Block 6 (ascending until the R154 cap is reached, then top set holds per R155).
- R71.** IF sets = 6–7 (heavy strength only) THEN rest = 150–240 seconds AND the first 1–2 sets of the lift are **ramp sets** at 60% then 70% of the working load (these are distinct from the session-level Warm-up block in R29). Working set load and rep distribution is governed by Block 6.
- R72.** IF athlete has provided a 1RM for the lift THEN use %1RM per Block 6.
- R73.** IF athlete has not provided a 1RM but has logged at least 2 working sets of that lift in-app THEN estimate 1RM using the Epley formula ($\text{weight} \times (1 + \text{reps} / 30)$) from the heaviest set, and apply Block 6 against the estimate.
- R74.** IF athlete has no 1RM and no logged history for that lift THEN use RPE-based loading per rule 75, and prompt the user to log perceived effort after each set.

R75. RPE fallback table:

Set / Rep Scheme	Target RPE
2–3 sets × 8–12 reps	RPE 6–7
4–5 sets × 4–6 reps	RPE 7–8
6–7 sets × 2–3 working reps	RPE 8–9 (working sets only)

- R76.** IF athlete is in week 1 of their first program AND has no 1RM data THEN week 1 prescribes RPE-based loading at moderate effort (RPE 6–7) **regardless of athlete level or program intent**. This rule overrides R74 in week 1 for all athletes (beginner, intermediate, and advanced). Starting week 2, the system uses logged data from week 1 to estimate 1RM via R73.
- R77.** IF exercise.fatigue_cost = high THEN rest must be ≥ 120 seconds.
- R78.** IF calf tendon training occurs THEN include heavy slow calf raises.
- R79.** IF knee tendon stress occurs THEN include isometric squat or Spanish squat.
- R80.** IF Achilles loading occurs THEN include eccentric or slow calf training.
- R81.** IF athlete performs high plyometric volume THEN increase tendon support work.
- R82.** IF session includes plyometrics AND athlete level = beginner AND plyometric intensity is NOT high THEN use 40–70 contacts.
- R83.** IF session includes plyometrics AND athlete level = intermediate AND plyometric intensity is NOT high THEN use 60–100 contacts.
- R84.** IF session includes plyometrics AND athlete level = advanced AND plyometric intensity is NOT high THEN use 80–120 contacts.
- R85.** IF plyometric intensity = high THEN total high-intensity contacts must be ≤ 25 regardless of athlete level.
- R86.** IF plyometric category = extensive THEN prioritize low-amplitude jumps.
- R87.** IF plyometric category = intensive THEN prioritize maximum-effort jumps.
- R88.** IF athlete performs maximal jumps THEN rest must be ≥ 2 minutes between sets.
- R89.** IF plyometric intensity increases THEN reduce plyometric volume.
- R90.** IF athlete performs a maximal CNS lift THEN the next training day must not contain another maximal CNS lift.
- R91.** IF athlete performs a heavy squat session THEN the next training day must not include maximal plyometrics.
- R92.** IF athlete has no prior session history (week 1 of program) THEN the first training week must not contain more than 1 maximal CNS session, regardless of weekly structure.

R93. IF training days per week ≥ 4 THEN maximal CNS lifts must be separated by at least one non-CNS-heavy training day (recovery, mobility, or accessory-focused session).

Block 5 — Progression, Deload, Blocks, Sport Templates

A training week is defined as 7 calendar days starting from the user's first completed workout. Adherence percentage = (completed sessions / prescribed sessions) within that 7-day window.

R94. IF weekly adherence $< 75\%$ THEN repeat training week and do not progress load or intensity.

R95. IF weekly adherence is $75\text{--}89\%$ AND athlete completes all prescribed reps THEN allow small progression.

R96. IF weekly adherence $\geq 90\%$ AND athlete completes all prescribed reps THEN progress training intensity.

R97. IF athlete fails prescribed reps THEN decrease load $5\text{--}10\%$.

R98. IF athlete misses workouts THEN do not progress training load unless weekly adherence remains $\geq 75\%$.

R99. IF a weekly program is generated THEN include at least one tendon loading exercise.

R100. IF tendon adaptation phase begins THEN progress from isometric $\rightarrow$ slow resistance $\rightarrow$ plyometric.

R101. IF athlete tolerates tendon loading THEN gradually increase plyometric intensity.

R102. IF sprint distance $\leq 30\text{m}$ THEN classify sprint as acceleration work.

R103. IF sprint distance $\geq 40\text{m}$ THEN classify sprint as max velocity work.

R104. IF a sprint repetition is performed THEN rest must be $\geq 2\text{--}3$ minutes.

R105. IF training week = deload week THEN reduce volume by $50\text{--}60\%$.

R106. IF training block length = 4 weeks THEN week 5 must be a deload week.

R107. IF a new block begins THEN increase training intensity gradually.

R108. IF strength block ends THEN begin power emphasis block.

R109. IF power block begins THEN reduce heavy lifting volume.

R110. IF athlete plateaus THEN vary exercise selection.

R111. IF athlete improves strength significantly THEN introduce more velocity-based work.

R112. IF primary goal = vertical jump THEN prioritize elastic and strength exercises.

R113. IF primary goal = speed THEN prioritize velocity and acceleration exercises.

R114. IF primary goal = strength THEN prioritize compound loadable lifts.

R115. IF primary goal = injury prevention THEN prioritize mobility and unilateral exercises.

R116. IF session intent = strength THEN $\geq 50\%$ of exercises must be strength intent.

R117. IF session intent = power THEN $\geq 50\%$ of exercises must be velocity or elastic intent.

R118. IF session intent = recovery THEN $\geq 70\%$ of exercises must be mobility or low-fatigue exercises.

R119. IF session intent = recovery THEN exclude high-CNS-cost exercises.

R120. IF athlete enters a power-emphasis training block THEN increase maximal jump exercises, EXCEPT where this conflicts with rules 90 or 91 (CNS recovery), in which case the maximal jump is rescheduled to the next eligible training day.

R121. IF session contains a bilateral squat pattern THEN accessory movement must include a unilateral leg exercise.

R122. IF a hip hinge movement is used THEN accessory must target hamstrings or glutes.

R123. IF an upper body push exercise is used THEN include an upper body pull exercise.

R124. IF athlete performs heavy squats THEN include posterior chain accessory.

- R125.** IF athlete performs sprint drills THEN include hip mobility work.
- R126.** IF athlete performs an Olympic lift THEN reduce total lower body strength volume.
- R127.** IF athlete marks a workout as complete THEN update the weekly completion percentage.
- R128.** IF athlete completes the final workout of the week THEN generate the next week's program.
- R129.** IF athlete records improvement metrics THEN store data for future plan generation.
- R130.** IF athlete consistently completes workouts THEN increase program difficulty gradually.
- R131.** IF athlete repeatedly skips workouts THEN reduce training volume.
- R132.** IF athlete program cycle ends THEN regenerate a new training block.

Weekly Structure Templates

- R133.** IF training days = 2 THEN weekly structure = Day 1 Full Body Strength / Day 2 Power + Speed.
- R134.** IF training days = 3 THEN weekly structure = Day 1 Strength / Day 2 Upper + Mobility / Day 3 Power + Speed.
- R135.** IF training days = 4 THEN weekly structure = Day 1 Lower Strength / Day 2 Upper Strength / Day 3 Power + Speed / Day 4 Recovery / Mobility.
- R136.** IF training days = 5 THEN weekly structure = Day 1 Lower Strength / Day 2 Upper Strength / Day 3 Power / Day 4 Speed / Day 5 Recovery.

Sport-Specific Requirements

- R137.** IF sport = basketball THEN weekly structure must include 1 jump or reactive plyometric session AND 1 change-of-direction session.
- R138.** IF sport = football THEN weekly structure must include 1 acceleration session.
- R139.** IF sport = soccer THEN weekly structure must include 1 change-of-direction session.
- R140.** IF sport = track & field THEN weekly structure must include 1 max velocity sprint session.

Day Type to Primary Block Mapping

- R141.** IF day type = Full Body Strength or Lower Strength THEN primary block = Main Lift.
- R142.** IF day type = Power + Speed or Power or Speed THEN primary block = Power.
- R143.** IF day type = Upper + Mobility THEN primary block = Main Lift or Accessory.
- R144.** IF day type = Recovery or Mobility THEN primary block = Recovery.
- R145.** IF day type = Lower Unilateral or Volume THEN primary block = Accessory.
- R146.** IF training block = strength THEN prioritize loadable exercises.

Note: A taper-block rule existed in v4 (formerly R147) but has been removed in v5. Taper logic requires a competition-date trigger that is not part of V1 onboarding. When competition-date capture is added to onboarding in a future version, taper rules can be reintroduced with proper triggers.

Block 6 — Per-Set Load Prescription (NEW in v4)

Purpose: Block 6 governs how an exercise's prescribed load is distributed across its individual sets. Prior versions used a single %1RM for an entire exercise (e.g., "3 × 5 @ 75% 1RM"). v4 replaces this with per-set ascending schemes so that Set 1, Set 2, and Set 3 each have their own exact load. The driver is the exercise's **Load type** field in the exercise database.

Load Type → Per-Set Scheme Mapping

Load Type	Rep Range (within-session descent)	Scheme	Increment	Example (3 sets)
heavy strength	5 → 3	Ascending	+5%	5 @ 75% / 4 @ 80% / 3 @ 85%
power	3 → 1	Ascending	+10%	3 @ 70% / 2 @ 80% / 1 @ 90%
speed strength	5 → 3	Ascending	+5%	5 @ 50% / 4 @ 55% / 3 @ 60%
strength speed	5 → 3	Ascending	+5%	5 @ 60% / 4 @ 65% / 3 @ 70%
hypertrophy	12 → 8	Ascending	+5%	12 @ 60% / 10 @ 65% / 8 @ 70%
endurance	12–15 (constant)	Straight	—	15 @ 50% / 15 @ 50% / 15 @ 50%
mobility	time / reps	N/A	—	bodyweight
prehab (tendon iso)	30–45s (constant)	Straight	—	30s / 30s / 30s

Block 6 Rules

- R148.** Each set in a user-facing workout must display its own exact load and reps. The app must NOT display a single %1RM or load value applied to the entire exercise.
- R149.** Per-set load scheme is determined by the exercise's **Load type** field, not by rep range or category alone.
- R150.** IF Load type = heavy strength OR hypertrophy OR speed strength OR strength speed THEN load scheme = ascending at +5% per working set.
- R151.** IF Load type = power THEN load scheme = ascending at +10% per working set (maximal intent justifies steeper jumps).
- R152.** IF Load type = endurance THEN load scheme = straight sets (constant load across all working sets).
- R153.** IF Load type = mobility OR prehab THEN no %1RM is applied; prescription is bodyweight, light fixed load, or time/reps only.
- R154.** Top set on ascending schemes is capped by athlete training level: **beginner** ≤ 80% 1RM, **intermediate** ≤ 87% 1RM, **advanced** ≤ 92% 1RM. The cap does NOT apply during a tested 1RM week and does NOT apply to the power Load type (which is intentionally maximal-intent at low reps).
- R155.** IF set count exceeds the number of distinct ascending steps available within the Load type's safe range THEN the top set is held (repeated) on subsequent sets rather than continuing to ascend. Example: heavy strength, intermediate (cap 87%), 5 working sets → 75% / 80% / 85% / 85% / 85%.
- R156.** Starting load for an ascending scheme = low end of the working set range for that Load type. The starting percentage may be raised within the range based on prior-week performance per Block 5 rules 95–98.
- R157.** IF sets = 6–7 (Load type = heavy strength only) THEN the first 1–2 sets are **ramp sets** at 60% then 70% of the working load (these are part of the lift itself, distinct from the session-level Warm-up block in R29), followed by ascending working sets that top out at the R154 cap. Example: advanced, 7 sets → 60% / 70% /

75% / 80% / 85% / 90% / 90%.

R158. IF the athlete has no 1RM and no logged history (per R74) THEN ascending schemes use RPE instead of %1RM: Set 1 = RPE 6, Set 2 = RPE 7, Set 3 = RPE 8 (capped at RPE 9 for advanced athletes only). Beginner cap = RPE 8. Intermediate cap = RPE 8.5.

R159. IF the exercise's **Loadable** field = false THEN display reps or time only; do NOT display a %1RM or load value, regardless of Load type.

R160. IF Load type = endurance (straight sets) AND set count > 3 THEN load remains constant across all sets but rest may be reduced by 10–15 seconds per set as a density progression option in later weeks.

R161. Within-session rep behavior for ascending Load types (Q1 = within-week descent): As load ascends across sets, reps descend across the prescribed rep range. The first set uses the high end of the range, the last working set uses the low end. Default rep distribution by Load type:

- **heavy strength (1–5)**, 3 sets: 5 / 3 / 1 OR 5 / 4 / 3 (advanced may use 5/3/1; beginner/intermediate use 5/4/3).
- **power (1–3)**, 3 sets: 3 / 2 / 1.
- **hypertrophy (8–12)**, 3 sets: 12 / 10 / 8.
- **speed strength / strength speed (3–5)**, 3 sets: 5 / 4 / 3.
- **endurance (12–15)**: reps stay constant across sets (straight set scheme; no descent).
- **prehab tendon isometrics (R66)**: duration stays constant across sets (e.g., 30s / 30s / 30s). Duration progresses week-over-week, not within a session.
- **mobility**: reps stay constant; no load progression.

For 4–5 set protocols where R155 holds the top set, the held sets repeat the lowest rep value (e.g., heavy strength 5 sets: 5 / 4 / 3 / 3 / 3).

R162. Per-set load values must be rounded to the nearest 5 lb (or 2.5 kg) for plate-loaded barbell exercises before display. Dumbbell exercises round to the nearest available dumbbell increment.

R163. IF Block 1 (Safety & Pain Overrides) caps an exercise's intensity at 80% (e.g., mild knee/back/shoulder pain rules) THEN the Block 6 ascending top set must respect that 80% cap, overriding R154.

R164. Rest tiebreaker: When multiple rules prescribe rest for the same exercise or transition (e.g., R69–R71 set-count rest, R77 fatigue_cost rest, R88 maximal jumps rest, R104 sprint rest), **use the longest required rest value**. Example: a maximal jump performed in a 4-set protocol triggers R70 (120–180s) and R88 (≥120s) and R104 if it qualifies as sprint work (≥120–180s) — final rest = 180s minimum.

Worked Examples

Exercise	Load Type	Sets × Reps	Per-Set Display
Back Squat (intermediate)	heavy strength	3 × 5–3	Set 1: 5 @ 75% Set 2: 4 @ 80% Set 3: 3 @ 85%
Back Squat (intermediate, 5 sets)	heavy strength	5 × 5–3	Set 1: 5 @ 75% Set 2: 4 @ 80% Set 3: 3 @ 85% Set 4: 3 @ 85% Set 5: 3 @ 85%
Power Clean (advanced)	power	3 × 3–1	Set 1: 3 @ 70% Set 2: 2 @ 80% Set 3: 1 @ 90%

DB Bench Press (intermediate)	hypertrophy	3 × 12–8	Set 1: 12 @ 60% Set 2: 10 @ 65% Set 3: 8 @ 70%
Goblet Squat (beginner)	endurance	2 × 15	Set 1: 15 @ 50% Set 2: 15 @ 50% (constant — straight sets)
Spanish Squat Iso (any level)	prehab	3 × 30s	Set 1: 30s hold Set 2: 30s hold Set 3: 30s hold (constant within session)
90/90 Hip Switch (any level)	mobility	2 × 8	Set 1: 8 reps Set 2: 8 reps (bodyweight)

Implementation note for developer: The per-set prescription engine should expose a function with this signature:

```
getPerSetPrescription(exercise, athlete, weekContext) → Array<{ setNumber, reps,
loadPercent, displayLoad }>
```

Inputs come from the exercise database (Load type, Loadable, etc.), the athlete profile (training level, 1RM, history), and the weekly context (week number, prior adherence, deload flag). Block 6 rules R148–R163 are the logic this function implements. This function runs *AFTER* Blocks 1–5 have produced the session plan; it does not change exercise selection, only how each selected exercise is displayed.

Implementation Confirmation Checklist

The developer should confirm these are implemented either as explicit rules or directly in code:

- Exact reps only on user-facing workouts (no ranges).
- Each set displays its own exact load (no single %1RM applied to entire exercise) — see Block 6.
- Distance exercises display distance, not time.
- Time-based exercises display time, not reps or distance.
- Rep/set exercises do not display as distance.
- Unilateral exercises display a "both sides" reminder.
- Maximum of 8 displayed exercises per workout, excluding grouped warm-up/cool-down movements if applicable.
- Bodyweight is supported as a load type.
- Exercise videos play in-app if possible.
- Load type field is required on every exercise in the database (Block 6 cannot run without it).
- Loadable field is supported and respected (R159) — exercises tagged Loadable=false display reps/time only.
- Per-set loads round to nearest 5 lb / 2.5 kg before display (R162).